

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12408648
Kind Code	B2
Date of Patent	September 09, 2025
Inventor(s)	Jalowiec; Thomas Walter

Method of trapping ticks

Abstract

Provided is a method of trapping ticks using a tick trap having a glue constructed to trap ticks touching the glue by sticking to the ticks, and a plant based scent formed from a part of a plant that attracts ticks to the glue, the method includes placing the trap on a location surface and using the tick attractant to attract ticks to the glue.

Inventors:	Jalowiec; Thomas Walter (Sauquoit, NY)
Applicant:	Tick Sticker LLC (Boca Raton, FL)
Family ID:	63245434
Assignee:	TICK STICKER LLC (Boca Raton, FL)
Appl. No.:	15/872697
Filed:	January 16, 2018

Prior Publication Data

Document Identifier	Publication Date
US 20180242568 A1	Aug. 30, 2018

Related U.S. Application Data

division parent-doc US 15715481 20170926 PENDING child-doc US 15872697
us-provisional-application US 62518849 20170613
us-provisional-application US 62529064 20170706
us-provisional-application US 62551658 20170829
us-provisional-application US 62463887 20170227

Publication Classification

Int. Cl.: A01M1/14 (20060101); A01M1/02 (20060101); A01M1/10 (20060101)

U.S. Cl.:

CPC A01M1/14 (20130101); A01M1/02 (20130101); A01M1/103 (20130101); A01M1/10 (20130101); A01M2200/011 (20130101); Y02A50/30 (20180101)

Field of Classification Search

CPC: A01M (1/00); A01M (1/02); A01M (1/023); A01M (1/10); A01M (1/103); A01M (1/14)

USPC: 43/107; 43/114; 43/121; 43/132.1

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2911756	12/1958	Geary	43/114	A01M 1/02
3094805	12/1962	Luck	43/131	A01M 25/004
3685199	12/1971	Bradshaw	43/114	A01M 1/145
3729858	12/1972	Bradshaw	43/114	A01M 1/14
3755958	12/1972	Bradshaw	43/114	A01M 1/02
3863384	12/1974	Weatherston	43/114	A01M 1/14
3913259	12/1974	Nishimura	43/114	A01M 1/02
4156321	12/1978	Capizzi	43/114	A01M 1/14
4161079	12/1978	Hill	43/114	A01M 23/005
4208828	12/1979	Hall	43/114	A01M 1/02
4815231	12/1988	McQueen	43/114	A01M 1/14
4933371	12/1989	Hink	N/A	N/A
4992268	12/1990	Landolt	424/407	A01M 1/02
5031354	12/1990	Olson	43/114	A01M 1/02
5119586	12/1991	Townsend	43/114	A01M 1/023
5123201	12/1991	Reiter	43/107	A01M 1/026
5189830	12/1992	Montemurro	43/121	A01M 1/023
5258176	12/1992	Keenan	424/84	N/A
5381557	12/1994	Luria	2/16	A01M 1/14
5454186	12/1994	Gang	43/114	A01M 1/02
5461822	12/1994	Green	43/107	A01M 1/02
5572825	12/1995	Gehret	229/120.11	A01M 1/02
5839221	12/1997	Ron	43/132.1	A01M 1/02
5907923	12/1998	Heath	43/107	A01M 1/02
5939062	12/1998	Heath	424/405	A01M 1/02
6138402	12/1999	Wotton	43/107	A01M 1/02
6155002	12/1999	Holder	43/114	A01M 1/14
6264939	12/2000	Light	424/84	A01M 1/02
6389740	12/2001	Perich	424/409	A01M 1/04
6415545	12/2001	Watanabe	N/A	N/A
6516558	12/2002	Lingren	43/107	A01M 1/02

6528049	12/2002	Light	424/84	A01M 1/02
6920716	12/2004	Kollars, Jr.	43/107	A01M 1/023
7073287	12/2005	Lau	43/107	A01M 1/02
7434351	12/2007	Bette	43/107	A01M 1/106
7856752	12/2009	Eilersen	43/114	A01M 23/005
8104223	12/2011	Rodriguez	43/114	A01M 1/14
8109035	12/2011	Bowden	43/107	A01M 1/106
8240081	12/2011	Cuellar Bernal	43/107	A01M 1/145
8402690	12/2012	Schneidmiller	43/121	A01M 1/103
8424239	12/2012	Gallo	43/112	A01M 1/223
8793927	12/2013	Winkler	43/107	A01M 1/14
8978290	12/2014	Wright	43/113	A01M 1/02
9012385	12/2014	Di Biase	508/463	C08F 220/68
9066508	12/2014	Aroniss	N/A	A01M 1/106
9307755	12/2015	Bassan, Jr.	N/A	A01M 1/2094
9681652	12/2016	Petersen	N/A	A01M 1/02
10098337	12/2017	Yans	N/A	A01M 1/14
2002/0132020	12/2001	Lillig	N/A	N/A
2004/0216366	12/2003	Ritchie	43/114	A01M 1/026
2005/0144831	12/2004	Knauf	43/107	A01M 1/02
2005/0238713	12/2004	Kollars, Jr.	424/466	A01M 1/023
2006/0016120	12/2005	Masters	43/114	A01M 1/023
2006/0150473	12/2005	Bette	N/A	N/A
2006/0198857	12/2005	Casagrande	424/405	A01M 1/02
2006/0283075	12/2005	Feldhege	43/114	A01M 1/02
2007/0004686	12/2006	Bengtsson	514/159	A01N 31/04
2007/0094915	12/2006	Plato	43/114	A01M 1/02
2008/0052982	12/2007	Windsor	43/114	A01M 1/14
2008/0086932	12/2007	Cook	N/A	N/A
2009/0071059	12/2008	Della Torre	43/114	A01M 1/145
2009/0223115	12/2008	Lang	43/114	A01M 1/023
2009/0260276	12/2008	Kirsch	N/A	N/A
2010/0154290	12/2009	Fisher	43/114	A01M 1/02
2012/0055075	12/2011	Winkler	43/114	A01M 1/02
2012/0124888	12/2011	Lemberg	43/107	A01M 1/14
2012/0285076	12/2011	Banfield	43/123	A01M 1/026
2013/0111802	12/2012	Oehlschlager	43/114	A01M 1/14
2013/0340315	12/2012	Gustafson	43/2	A01M 31/06
2014/0105952	12/2013	Gilbert	N/A	N/A
2014/0242199	12/2013	Manhas	N/A	N/A
2014/0290123	12/2013	Duff	43/114	A01M 1/02
2016/0066555	12/2015	Yans	43/114	A01M 1/14
2016/0073622	12/2015	Jany	43/131	A01N 25/006
2016/0120164	12/2015	Li	43/114	A01M 1/14
2017/0094960	12/2016	Sasaki	N/A	H02S 40/38
2018/0242567	12/2017	Jalowiec	N/A	A01M 1/10
2019/0000062	12/2018	Yans	N/A	A01M 1/14
2019/0327952	12/2018	Jalowiec	N/A	A01M 1/14

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
1079636	12/1996	CN	N/A
103314938	12/2012	CN	N/A
106818674	12/2016	CN	N/A
3205736	12/2000	JP	N/A
3878366	12/2006	JP	N/A
3163352	12/2009	JP	N/A
3163352	12/2009	JP	N/A
5 121228	12/2012	JP	N/A
2003 0026067	12/2002	KR	N/A

OTHER PUBLICATIONS

Translation of CN 106818674 (Year: 2017). cited by examiner

Translation of CN 103314938 (Year: 2013). cited by examiner

Tick Catcher Sheet, Japan Made, <https://japanmade.co.jp/product/tick-catcher-sheet/>, 2017, pp. 1-7. cited by applicant

Tick, Wikipedia, <https://en.wikipedia.org/wiki/Tick>, 2018, pp. 1-10. cited by applicant

Mites that Bug People, <https://content.ces.ncsu.edu/mites-that-bug-people>, Jun. 1, 2015, pp. 1-8. cited by applicant

Mite, Wikipedia, <https://en.wikipedia.org/wiki/Mite>, 2018, pp. 1-9. cited by applicant

“DIY Homemade Natural Tick Repellent Recipes,” www.tickbites.net/diy-homemade-natural-tick-repellent-recipes/, Aug. 2017, pp. 1-12. cited by applicant

Grunert, “Natural Home Remedy Tick Repellent,”

www.herbs.lovetoknow.com/Natural_Home_Remedy_Tick_Repellent, Aug. 2017, pp. 1-3. cited by applicant

Bauer, “5 Natural Bug Repellents that Actually Work,” www.cheatsheet.com/life/5-natural-bug-repellents-that-actually-work.html, Aug. 2017, pp. 1-7. cited by applicant

Fabio, “Repellent effect of sweet basil compounds on *Ixodes ricinus* ticks,”

www.ncib.nlm.nih.gov/pubmed/18670893, Aug. 2008, p. 1. cited by applicant

“Plants that help control ticks,” May 2013, www.masterplanlandscapedesignpa.com, pp. 1-6. cited by applicant

Catchmaster fold up trap, <http://www.catchmaster.com/wpcproduct/mouse-insect-glue-boards/>, pp. 1-2, 2010. cited by applicant

“Scientists Says Popular Shrub Linked to Lyme Disease,” Pest Control Technology, Oct. 3, 2017, pp. 1-3, <http://www.pctonline.com/article/scientists-poular-shrub-lyme-disease-tick/?isid=MzQyNDQ2Mw%3d%3d>. cited by applicant

Invitation to Pay Additional Fees and where applicable protest fee, issued in PCT/US18/13806,

Apr. 23, 2018, pp. 1-12. cited by applicant

Protest and Response to Invitation to Pay Additional Fees and where applicable protest fee, issued in PCT/US18/13806, Apr. 26, 2018, pp. 1-4. cited by applicant

International Search Report issued in PCT/US18/13806, Jun. 21, 2018, pp. 1-13. cited by applicant

Final Office Action issued in Tick Sticker trademark registration No. 87868112, Mar. 8, 2019. cited by applicant

by applicant

Buggy Beds, Jun. 21, 2019, pp. 1-5, <https://buggybeds.com/collections/all>. cited by applicant

Pest-Control Tokyo Kikaku Hanbai Co., Ltd., Jun. 21, 2019, Tick Trap Sheet, p. 1. cited by applicant

Response to Written Opinion filed in PCT/US18/13806, Mar. 4, 2019, pp. 1-42. cited by applicant

Written Opinion issued in PCT/US18/13806, Jan. 28, 2019, pp. 1-9. cited by applicant
International Search Report and Written Opinion issued in PCT/US2019/056424, Mar. 18, 2020, pp. 1-25. cited by applicant
Bayala et al. "Chemical Composition, Antioxidant, Anti-Inflammatory and Anti-Proliferative Activities of Essential Oils of Plants from Burkina Faso" PLOS ONE. Mar. 24, 2014 (Mar. 24, 2014) vol. 9, p. 1-11; p. 1, abstract. cited by applicant
Invitation issued in PCT/US19/56424, Dec. 17, 2019, pp. 1-2. cited by applicant
Protest filed in PCT/US19/56424, Jan. 10, 2020, pp. 1-3. cited by applicant
Notification issued in PCT/US19/56424, Feb. 12, 2020, pp. 1-3. cited by applicant
Request for Supervisory Review filed in PCT/US19/56424, Jan. 17, 2020, pp. 1-7. cited by applicant
European Search Report issued in corresponding EP patent application 18 75 8103, Nov. 25, 2020, pp. 1-6. cited by applicant
Office Action issued in corresponding Indian Patent Application Serial No. 201937037173, Sep. 7, 2021, pp. 1-11. cited by applicant
Office Action issued in corresponding Chinese Patent Application Serial No. 201880014244.1, Mar. 23, 2021, pp. 1-9, including translation. cited by applicant

Primary Examiner: Ark; Darren W

Attorney, Agent or Firm: Melcher Patent Law PLLC

Background/Summary

FIELD OF THE INVENTION

(1) The invention relates to a method of trapping ticks.

BACKGROUND OF THE INVENTION

(2) Lyme disease is carried by ticks. Lyme disease can cause death in pets, especially dogs. Lyme disease can cause severe problems in humans. It is estimated that about 300,000 people a year in the U.S. and 65,000 people a year in Europe become infected with Lyme disease.

(3) Ticks are also carriers of diseases such as Rocky Mountain Spotted fever, anaplasmosis, Q fever, hemorrhagic fever, Powassan virus, and encephalitis.

(4) One attempted solution is to spray various poisons to kill the ticks. Spraying poisons, however, has an adverse effect on the environment and is potentially dangerous to humans, mammals and beneficial insects such as bees.

(5) U.S. Pat. No. 5,189,830 discloses a tick trap having a raised base having inclined walls at each end, an upper portion having sticky tick trapping surface and an upwardly extending box-like portion having a plurality of holes around the lower surface thereof, a rectangular rain guard extending outwardly from the box at a spaced distance from the surface of the base and a hinged lid over the open end of the box. The box includes a carbon dioxide canister which is periodically activated by batteries to emit carbon dioxide from the portholes about the box to attract ticks onto the sticky surface (glue) where they are destroyed. This product and method are not efficient and simply are not effective.

SUMMARY OF THE INVENTION

(6) An objective of the invention is to provide an efficient tick trap and method that does not require batteries, can be utilized continuously, and have directional attraction of ticks.

(7) Another objective of the invention is method of providing a tick reduced safe area where pets and humans can enjoy the outdoors with reduced fear of being bitten by a tick.

(8) The objectives of the invention and other objectives can be obtained by a method of trapping ticks comprising: placing a tick trap at a location to trap ticks, the tick trap comprising: a support; a glue on the support, the glue is constructed to trap ticks touching the glue by sticking to the ticks; and a tick attractant comprising a plant based scent formed from a part of a plant; and, the plant based scent attracting ticks to the glue so that ticks contact the glue and become trapped by the glue.

(9) The objectives of the invention can also be obtained by a method of trapping ticks comprising: folding a support to provide a housing having at least one side opening and inside surfaces defining a protective enclosure, a glue being on at least one inside surface, the glue is constructed to trap ticks touching the glue by sticking to the ticks, a tick attractant is incorporated in the glue, is on the glue, is on the support, or is on the housing and the tick attractant comprising a plant based scent formed from a part of a plant; placing the housing on a location surface to trap ticks; and the plant based scent attracting ticks so that ticks walk from the location surface through the side opening directly into the protective enclosure and contact the glue, wherein the glue traps the ticks.

(10) The objectives of the invention can further be obtained by a method of trapping ticks comprising: providing a housing having at least one side opening and inside surfaces defining a protective enclosure; providing a support constructed to be removably inserted into the housing, a glue being on support, the glue is constructed to trap ticks touching the glue by sticking to the ticks, a tick attractant is incorporated in the glue, is on the glue, is on the support, or is on the housing, and the tick attractant comprising a plant based scent formed from a part of a plant; placing the housing on a location surface to trap ticks; and the plant based scent attracting ticks so that ticks walk from the location surface through the side opening directly into the protective enclosure and contact the glue, wherein the glue traps the ticks.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 illustrates an example of a tick trap.

(2) FIG. 2 illustrates a top view of an example of a glue trap.

(3) FIG. 3 illustrates a side view of an example of a tick trapping chamber containing a glue trap.

(4) FIG. 4 illustrates a side view of an example of a tick trapping chamber containing a glue trap.

(5) FIG. 5 illustrates a view of an opening in an example of a tick trapping chamber in the shape of a cone of FIG. 1.

(6) FIG. 6 illustrates a view of an opening in an example of a tick trapping chamber in a cone of FIG. 3 and tube of FIG. 4 each having a flat bottom for containing a glue trap.

(7) FIG. 7 illustrates a tick trapping chamber connected to a tick capture structure comprising a second chamber.

(8) FIG. 8 illustrates tick reduced area surrounded by tick traps.

(9) FIGS. 9A, 9B and 9C illustrate examples of different covers.

(10) FIGS. 10A and 10B illustrate examples of different angles the carbon dioxide exits the chamber.

(11) FIG. 11 illustrates an example of a system utilizing a large canister connected to numerous tick trapping chambers to provide a tick reduced zone.

(12) FIGS. 12A and 12B illustrate different tick tapping chambers that disperse the carbon dioxide in all directions.

(13) FIG. 12C illustrates multiple openings into the tick trap.

(14) FIGS. 13A illustrates an outside view and 13B illustrates an inside view of an example of a tick trap.

(15) FIGS. 14A illustrates an outside view and 14B illustrates an inside view of an example of a

tick trap.

(16) FIG. 15 illustrates a modified electrical insect shocker.

(17) FIG. 16A illustrates a foldable tick trap in an unfolded position. FIG. 16B illustrates a packaged attractant. FIG. 16C illustrates the unfolded tick trap having the attractant applied. FIG. 16D illustrates a folded tick trap. FIG. 16E illustrates a folded tick trap.

DETAILED DESCRIPTION OF THE INVENTION

(18) The invention will be explained with reference to the attached non-limiting FIGS. 1-16.

(19) As shown in FIG. 1, the tick trap 1 comprises a canister 2 containing carbon dioxide 40, a canister connector 7 constructed to connect to the canister 2, an optional pressure gauge 4 for measuring the pressure within the canister 2, an optional shut off valve 6, a mechanical metering valve 8 constructed to regulate the flow of carbon dioxide 40 from the canister 2, an optional bubbler 10 through which the carbon dioxide 40 flows in the form of bubbles to measure the amount of flow, a tube 12 connecting the bubbler 10 to a tick trapping chamber 16 defined by a walled structure 14 having an opening 18, and a tick capture structure 19.

(20) Canisters 2 for containing carbon dioxide are now well known. Examples of canisters 2 are paintball CO2 canisters, CO2 cartridges, carbonators for soda machines, and conventional canisters, such as those sold by www.airgas.com. Any suitable canister 2 can be utilized. In general, the larger the capacity the greater the number of tick trapping chambers 16 that can be connected to the same canister 2 and the longer the time the tick trap 1 can be run. Preferably, the canister 2 contains a valve 3 so that if the canister connector 7 is not present on the canister 2 the valve 3 prevents release of the carbon dioxide 40 from the canister 2. Most of the paintball canisters 2 and other canisters 2 have such a valve 3 already mounted on the canister 3.

(21) Canister connectors 7 are now well known. Any suitable connector 7 for connecting the canister 2 to the system can be utilized.

(22) Tubing between the canister 2 and the mechanical metering valve 8 should high pressure tubing suitable for use with pressurized carbon dioxide 40. Such high pressure tubing is now well known and any suitable high pressure tubing can be used. Examples of suitable high pressure tubing include plastic or steel tubing. The tubing to transport the carbon dioxide 40 flowing from the mechanical metering valve 8 can be any suitable low pressure tubing, such as flexible or rigid aquarium tubing, for example acrylic or vinyl tubing. The size of the tubing will depend on the size of the connectors for each of the structures in the system. One skilled in the art will easily be able to select a suitable size tubing, for example, 1/8, 1/4, 5/16, 3/8, 1/2, 3/4, or 1 inch diameter tubing.

(23) The supply of carbon dioxide 40 from the canister 2 to the tick trapping chamber 16 can be conducted by modifying known aquarium carbon dioxide systems. Aquariums use a mechanical needle valve (metering valve) and bubbler attached to a canister containing carbon dioxide to supply carbon dioxide to the water for plant growth. These known carbon dioxide systems can now be modified for use in the claimed tick trap 1.

(24) A pressure gauge 4 to identify the amount of carbon dioxide 40 left in the canister 2 can be used. Pressure gauges are now well known and any suitable pressure gauge 4 can be utilized. The pressure gauge should be connected between the canister 2 and the mechanical metering valve 8.

(25) An optional shut off valve 6 can be connected to the canister 2 to shut off the carbon dioxide 40 flow without having to change the setting on the mechanical metering valve 8.

(26) The mechanical metering valve 8 is constructed to provide a low flow of carbon dioxide 40 from the canister 2. For outside the U.S., examples of suitable mechanical metering valves 8 include those sold by WYIN, such as WYIN W00-01C, WYIN JIS M22-14 mechanical needle valve to adjust the amount of carbon dioxide supplied from the canister. The connectors on these metering valves can be used in the U.S. by converting the size of the connector using adapters. Mechanical metering valves 8 conventionally used to control the flow rate of carbon dioxide for aquarium use can be utilized in the present invention.

(27) A bubbler 10 can be connected to the mechanical metering valve 8 using low pressure tubing

to visually measure the flow rate of the carbon dioxide **40**. Bubbblers **10** are now well known and any suitable bubbler can be utilized. The bubbler **10** comprises a vessel that is either clear or has a clear window that is filled with a liquid and has an input connector on the bottom and an exit connector on the top. The carbon dioxide **40** flows through the liquid in the form of bubbles. The greater the number of bubble per minute identifies a greater flow rate. When using a bubbler **10**, the flow rate can be any desired rate based on the bubbles per second. The flow rate can be adjusted to account for the surroundings and the type of ticks. For example, a higher flow rate can be first utilized to attract ticks, and then subsequently reduced by trial and error to determine a minimum flow rate to still attract the ticks to maximize the run time of the canister **2**. A preferred flow rate is the minimum to attract ticks to maximize the length of use for the canister **2**.

(28) The tick trapping chamber **16** can be connected to the mechanical metering valve **8** directly or through the optional bubbler **10**. The tick trapping **16** is defined by a walled structure **14**. The tick trapping chamber **16** comprises an opening **18** that allows the carbon dioxide **40** to exit the tick trapping chamber **16**. The tick trapping chamber **16** is in communication with a tick capture structure **19**. The tick capture structure **19** captures ticks that enter or come near the tick trapping chamber **16**. The tick capture structure **19** can be contained within the tick trapping chamber **16** or be near the tick trapping chamber **16**.

(29) The tick trapping chamber **16** can have any desired shape. Examples of shapes include square, rectangular, triangular, circular, cone, or any other desired shape. Preferably, the tick trapping chamber **16** has the shape of an at least partially tubular, FIG. **4**, or at least partially cone shaped FIG. **3**. The chamber **16** and opening **18** can be constructed to direct the carbon dioxide **40** flow in a desired direction and/or amount of dispersion. In FIG. **10A**, a top view of a chamber **16** in which the walls of the chamber **16** are not angled due to at least the partially tubular shape of the walled structure **14**, the carbon dioxide **40** flow is minimally dispersed in a mostly a singular direction. In FIG. **10B**, a top view of a chamber **16** in which the walls of the chamber **16** are angled due to the at least partially cone shape of the walled structure **14**, the carbon dioxide **40** flow is more dispersed. The angle of the chamber **16** can be any desired angle to provide the desired dispersion. Preferably, there are no openings on the side walls of the walled structure **14**, i.e. the opening **18** is preferably in a mostly singular direction as show in FIGS. **8,10A** and **10B**.

(30) The tick capture structure **19** can be any suitable structure for capturing ticks. Examples of suitable tick capture structures **19** include a sticky surface **20** or a container (second chamber) **30**. The sticky surface can be formed from any suitable sticky material, such as those used in fly traps and glue traps **22**.

(31) The tick capture structure **19** can be a sticky surface **20** (glue) within the tick trapping chamber **16** as shown in FIG. **1**. The tick trapping chamber **16** be sized to contain a commercially sold glue trap **22** as the tick capture structure **19** so that the user can easily purchase and change the glue trap **22** as desired. Glue traps **22** are commercially sold at stores such as Walmart, Home Depot, Lowes, and other stores. Examples of glue traps **22** include Harris, Dead End, Real Kill, Catchmaster, RatGuard, MouseGuard, and similar brands. Examples of glue traps **22** can be from 2-4 inches wide to 4-8 inches long. However, any size glue trap **22** can be used that fits within the tick trapping chamber **16**. The tick trapping chamber **16** can be sized to fit any desired glue trap **22**. Glue traps **22** for animals can be used to trap the ticks. A glue board refill, such as Zap n trap, which is 18 inches by 10 inches can be cut to a desired size and utilized as the glue trap **22**. The glue traps can be replaceable.

(32) The tick capture structure **19** can be a container as shown in FIG. **7** from which the ticks have difficulty escaping. FIG. **7** illustrates a second chamber **30** defined by a second walled structure **34**. The second walled structure **34** can be a net, bag, jar, or other suitable container for capturing ticks. A sliding surface **32** is constructed so that when ticks contact the sliding surface **32** the ticks slide into the second chamber **30**. The sliding surface **32** and walled structure **34** are constructed so that it is more difficult for the ticks to exit the chamber **30** than for the ticks to enter the chamber **30**.

(33) The opening **18** can be numerous small openings, FIG. **9A**, a large opening covered by a screen FIG. **9B**, a slot FIG. **9C**, or any desired opening sized to allow ticks to enter but not larger insects, mammals, and reptiles to prevent them from being undesirably trapped. The opening **18** can be formed by a removable cover **24** that can be removed by a user to access the chamber **16**. Thus, the cover **24** can be a screen as shown in FIG. **9B**, the cover **24** can have holes as shown in FIG. **9A**, or the cover can have a slot as shown in FIG. **9C**. Any desired opening shape can be utilized.

(34) Multiple tick tapping chambers **16** can be connected to the same canister **2** as shown in FIG. **11**. Preferably, each tick trapping chamber **16** has an associated tick capture structure **19**, bubbler **10**, and mechanical metering valve **8**. Examples of commercially available multiple micrometering valves **8** include WYIN W00-02 dual head aluminum bubble counter with check valve to prevent return flow from pressure changes, WYIN W02-04 (four valves), WYIN W02-03 (three valves), and WYIN W01-05 (two valves). The tick trapping chambers **16** can be connected to their associated mechanical metering valve **8** by low pressure tubing of any desired length. Thus, the tick trapping chambers **16** can be moved around a desired area to provide a tick reduced area **50** shown in FIG. **11** with overlapping clouds of carbon dioxide **40** or by providing a perimeter as shown in FIG. **8**. If desired multiple tick trapping chambers **16** can be connected to a single metering valve **8** as shown in FIG. **12A** to provide a cloud of carbon dioxide **40** around the chambers **16**. FIGS. **12B** and **12A** illustrate the top and side views of a tick trapping chamber **16** in which the cloud of carbon dioxide is dispersed around the entire tick trapping chamber **16**.

(35) Preferably, the metering valve **8** is a fixed metering valve that cannot be adjusted by the user. For example, the metering valve **8** can be factory adjusted and then locked so that the user cannot adjust the flow rate of the carbon dioxide. Alternatively, the metering valve **8** can be constructed to be non-adjustable.

(36) A dwelling, such as a home, commercial building, or other structure, can be protected by connecting multiple tick traps throughout the dwelling. The multiple tick traps can be connected to a single large canister of carbon dioxide or multiple canisters of carbon dioxide.

(37) The tick traps **1** can be utilized inside dwellings, buildings and any other desired place since the traps **1** can be non-toxic to humans and pets. Thus, tick traps **1** can be placed near doorways to protect dwellings from ticks. For example, a tick trap can be placed in a closet where cloths are kept so that ticks brought in on the cloths can be trapped instead of feeding on mammals inside the dwelling. Preferably, the traps are free of ingredients that are toxic, poisonous, or harmful to humans or pets.

(38) When placing the traps **1** outside, preferably the chamber **16** (also referred to as a protective enclosure) is positioned so that ticks can walk from the ground or grass through the side opening **18** and directly into the chamber **16** to contact the glue **20**, **22** at the bottom of the housing **14** or support **100**. Grass can be placed at or in the side opening **18** entrance of the chamber **16**.

(39) The carbon dioxide **40** can attract ticks from 50 to 75 feet from the tick trapping chamber **16**. Thus, a tick reduced area **50** can be formed by providing a plurality of tick trapping chambers **16** around a desired tick reduced area. The tick trapping chambers **16** should be no more than 50 feet from one another, preferably no more than 40 feet from one another, and most preferably no more than 30 feet from one another to provide a swath of carbon dioxide **40** around the tick reduced area **50**, as shown in FIG. **8**.

(40) Since mosquitoes, and other insects, are attracted to carbon dioxide, the present invention can be utilized for trapping insects other than ticks. For trapping mosquitoes, the second walled structure can be constructed and sized accordingly. Fleas may also be trapped by the different tick traps disclosed herein.

(41) FIGS. **13A** and **13B** illustrate an example of a self contained tick trap **60** having a housing **14** defining the tick trapping chamber **16** and openings **18** in the housing through which the ticks can enter the chamber **16**. The canister **2** is located above the chamber **16**. Sticky surfaces **20** (glue) can

be present in the chamber **16**. The trap **60** can have a handle **66** for carrying. The trap **60** can also have an additional attractant **64** as described below. The glue **20** can in form of a removable glue trap **22** that can be replaced by the user so that the housing **14** can be reused. FIG. **12C** also illustrates the use of multiple openings **18** into the tick trapping chamber **16**.

(42) Another embodiment of the invention relates to trapping ticks using attractants **64** in place of the carbon dioxide attractant described herein as shown in FIGS. **14A** and **14B** or in addition to the carbon dioxide attractant as shown in FIGS. **13A** and **13B**.

(43) A particularly preferred embodiment of the invention is the use of a plant based tick attractant in combination with a glue on a support. The term support includes any material to which the glue can be applied, including the housings, glue traps, and foldable traps described herein.

(44) FIGS. **14A** and **14B** illustrate a preferred example of a tick trap **140** containing a tick attractant **64**. The tick trap **140** has a housing **14** defining the tick trapping chamber **16** and openings **18** in the housing through which the ticks can enter the chamber **16**. Sticky surfaces **20** (glue) can be present in the chamber **16**. The trap **140** can have a handle **66** for carrying. Preferably the attractant **64** is not harmful to humans or other mammals. The attractant **64** can be included in the glue formulation **20** and/or added to the trap as shown at **64**. The trap **140** preferably has housing **14** to protect the glue formulation **20** from the environment, especially rain. The glue **20** can in form of a removable glue trap **22** that can be replaced by the user so that the housing **14** can be reused.

(45) The attractant **64** is preferably a natural plant scent, such as a tree scent or bush scent. The tree scent can be formed from an evergreen tree or deciduous tree as desired. Preferred tree scents are cedar and apple. Preferred bush scents are basil and peppermint, with basil being most preferred. The plant scent can be formed by grinding parts of the plant and using the ground up parts as an attractant in the glue and/or near the glue. The plant sent can also be formed by soaking a part of the plant, such as the leaves, bark and/or wood, in a solvent, such as water, to extract the plant scent from the plant parts. Any suitable solvent for dissolving or dispersing the plant scent from the plant can be utilized, such as hydrocarbons, alcohols, water, alkoxides, etc. The plant scent can be formed from one or more plants that grow in the area where the traps will be placed. Cedar and basil have been used to repel insects. Surprisingly, we have found that cedar and basil are effective attractant for ticks. Furthermore, essential oils of the plant can also be utilized as the plant based scent.

(46) Preferably the attractant **64** is a plant based scent formed from a commonly consumed food commodity, such as a food based plant (edible plant). Non-limiting examples of such a food based plant scent are basil and apple, which are commonly consumed food commodities. The U.S. Environmental Protection Agency exempts food based plants from regulation in insect traps. Thus, preferably, the plant based scent is a food based scent that the FDA exempts from the FIFRA Registration. For example, preferred plant based scents for use as the attractant **64** meet the requirements under 40 C.F.R. Section 180.950 so that the plant based scents are exempted from the requirement of a tolerance under FFDCA section **408**. Preferably, the entire trap is exempted from the requirement of a tolerance under FFDCA section **408**.

(47) To protect an area, the tick traps **140** can be placed around the perimeter, such as from 10 to 75 feet apart. The tick traps **140** can be used and placed in the same manner as the carbon dioxide tick traps **1** described herein. The tick traps **140** can be set in any room of a home, for example in where cloths are stored to attract and trap ticks that drop from cloths worn outside.

(48) Electric shocking traps **80** typically utilize a light **82** to attract insects to the electrical shocking elements **84** to electrocute and kill the insects. As shown in FIG. **15**, the electric traps **80** can now be modified by attaching a tube **12** to supply carbon dioxide to the electric shocking trap **80** to provide an additional attractant, carbon dioxide, to the trap **80**. Alternatively, the attractants **64** described herein can be used in addition to the carbon dioxide or in place of the carbon dioxide.

EXAMPLE

(49) 19 glue traps were set up with different scents applied to them. The scents were made with an

essential oil and water mixture and then sprayed liberally on the glue traps. The scents tested were ginger, vetiver, clove, spearmint, patchouli, grapefruit, lemongrass, nutmeg, lavender, basil, tea tree, ylang ylang, bergamot, eucalyptus, wintergreen, peppermint, frankincense, cypress, and cinnamon. These were tested many times over a 3 month period from March to May. Basil was the most successful at attracting ticks Cedar was also tested by soaking cedar branches and leaves in water. Cedar was successful at attracting ticks on numerous occasions. The testing area was in a mostly wooded area with some brush cover in Sauquoit, NY during the month of May. On one four day testing period rain occurred on two days, but basil and cedar were successful at attracting ticks. Weather conditions during testing varied between warm, cold, rainy, overcast, and sunny.

(50) During that three-month time period, the following attractants in essential oil form did not adequately attract ticks in Sauquoit, New York, during the month of May: ginger, vetiver, clove, spearmint, datchouli, grapefruit, lemongrass, nutmeg, lavender, tea tree, ylang ylang, bergamot, eucalyptus, wintergreen, frankincense, and cypress. These plant based scents may work for other locations and/or times of the year.

(51) Another example of a plant based scent can be made from crab apple leaves. When using a scent made from crab apple leaves in Sauquoit, New York, during the month of June, one glue trap trapped 10 ticks in less than 24 hours. Without being bound by any theory, I believe that the ticks are attracted to plants where mammals eat or forage. For example, deer are carriers of tick borne diseases such as Lyme disease. Deer like to eat the crab apples and leaves of the crab apple tree. Thus, I believe the ticks are attracted to the crab apple tree and wait to fall onto a deer eating the crab apples or leaves. During this same time, cedar scent was tested and collected 9 ticks in less than 24 hours. The term apple tree includes all forms of apple trees, such as crab apple trees.

(52) Each of TomCat, RealKill, Catchmaster Max-Catch, and Catchmaster Roach & Insect glue traps were tested with basil leaves, crab apple leaves and cedar branches. The testing area was in a mostly wooded area with some brush cover in Sauquoit, NY during the months of May-July 2017. Basil leaves caught 3-4 ticks/24 hours. Crab apple leaves caught 3-10 ticks/24 hours. Cedar branches caught 1-6 ticks/24 hours.

(53) Preferably, the traps having attractants and glue are free of ingredients that are toxic, poisonous, or harmful to humans or mammals, such as dogs, rabbits, hamsters, and cats. The glue is preferably inert, non-irritating and non-harmful (non-hazardous) to humans and mammals. The traps having attractants and glue can be made environmentally friendly and/or biodegradable, so the entire trap can be thrown away. Preferably, the trap is free of materials that are not biodegradable or are harmful to the environment.

(54) Glues that are non-toxic, non-harmful, and non-irritating to humans and mammals are now well-known. Any commercially available glue that is non-toxic, non-harmful and non-irritating to humans and mammals can be utilized in the present invention. Examples of such glues are commercially available from Catchmaster, Tomcat, D-CON, and Scotts.

(55) The housing of the tick trap is preferably formed from a biodegradable material. Even more preferable, the material is water-resistant. Suitable materials are now well-known. For example, biodegradable, water-resistant packaging materials can be utilized, such as those commercially sold by www.begreenpackagingstore.com, www.packagingzone.net, www.polymateltld.com, www.nanotechindustries.com, etc. In this regard, the trap preferable is formed from a cellulosic material, such as paper, that can have a water repellant coating thereon. For example, a GreenCoat (Nanotech Industries Inc.) can be used to provide good water resistance to cellulose material. Microorganisms can decompose these materials and then the material can be converted into carbon dioxide, water and biomass while composting in wet soil.

(56) When the biodegradable trap can be simply thrown away with the trapped ticks contained therein without harming the environment. If the trap will be used outside, the material selected is preferably water resistant so that the trap will resist degradation for at least a week, and preferable up to month when exposed to moisture or rain. If the trap is mistakenly left in the environment, the

trap can decompose. In this manner, the trap can be utilized in the environment for a week without degrading and then simply discarded with the trapped ticks contained therein or allowed to decompose in place.

(57) FIGS. **16A-16E** illustrate an example of a preferred foldable, throw away trap according to the present invention. The foldable trap comprises a support **100** having the glue **20** applied to the support **100**. The tick attractant **64** can be applied to the glue **20** or to the support **100** by the user. The support **100** can comprise a tab **102** and slot **104** sized to receive the tab **102**. When the support is folded to form a housing, as shown in FIGS. **16 D** and **16E**, the tab **102** is inserted into the slot **104** to form a chamber **16** and opening **64**. The foldable trap functions similar to commercially available foldable glue boards, such as those from Catchmaster, d-CON, or Tomcat, that can be folded to provide a housing. The conventional foldable traps can be utilized by modifying them to include the present tick attractant **64** and optionally by replacing the support **100** material with a biodegradable, moisture resistant material. Preferably, the packaging material holding multiple or single traps and the tick attractant **64** is also formed from a biodegradable material. The tick attractant **64** can be provided in a separate container **106** if desired. The user can remove the tick attractant **64** from the container **106** and place the tick attractant **64** on the glue **20** or support **100** and use the trap in open form, FIG. **16C** or the support **100** can be folded to provide an enclosure, **16D** and **16E**. The open form can be used under a bed or in a closet and the enclosed, folded-form can be used outside or where humans or mammals may be exposed to the trap. Preferably, the tick attractant **64** is the plant based attractant described herein.

(58) The biodegradable tick trap has many advantages over conventional traps. For example, the traps can be folded to form a protective enclosure, the plant based attractant installed in the enclosure, and then placed in the environment and left there. After trapping ticks and being exposed to the environment, such as moisture, the trap will eventually break down to create soil and the ticks will die after being trapped in the glue. Therefore, the traps do not even have to be removed. A large area can inexpensively be covered with no harm to the environment.

(59) Any of the traps described herein, and especially the biodegradable traps, can be colored to blend into the environment they will be placed, such as green for foliage, brown for dirt, or any form of camouflage so that the traps are not unsightly.

(60) During Aug. 17-24, 2017, the paper foldup traps Catchmaster were tested by adding plant based scents based on essential oils of basil, apple and cedar. The weather varied between 60-80 degrees F., and from sunny to rainy. The testing area was in a mostly wooded area with some brush cover in Sauquoit, NY during the months of May-July 2017. Basil trapped 6 ticks. Apple trapped 2 ticks. Cedar trapped 1 tick. Buck lure and doe lure were also tested but did not attract any ticks.

(61) On Sep. 22, 2017, four tick traps like those shown in FIGS. **14A-B** were set in a grassy area in Sauquoit, NY. Two of the traps utilized a combination of apple and apple seed essential oil and two of the traps utilized basil essential oil as the attractants. In less than 24 hours, each of the traps captured 5 ticks each on the glue. This result was very surprising in view of basil being widely known as a tick repellent. Merely placing traps without attractants and traps with other scents in the same area did not result in any ticks being trapped. Applicant has unexpectedly found that basil and other plant based scents are effective tick attractants, not repellents.

(62) The present invention provides many advantages over previously tried tick traps. I could not locate any tick traps in the major stores, including Walmart, Lowe's and Home Depot. Tick borne diseases are a major problem. The present invention solves this problem by using novel methods and devices. The plant based scent in combination with a glue trap provides an environmentally and animal friendly solution to the tick problem.

(63) While particular embodiments of the present invention have been illustrated and described, it would be obvious to those skilled in the art that various other changes and modifications can be made without departing from the spirit and scope of the invention. It is therefore intended to cover

in the appended claims all such changes and modifications that are within the scope of this invention.

Claims

1. A method of trapping ticks comprising: placing a tick trap at a location to trap ticks, the tick trap comprising: a support; a glue on the support, the glue is constructed to trap ticks touching the glue by sticking to the ticks; and a tick attractant comprising a plant based scent formed from basil; and, the plant based scent attracting ticks to the glue so that ticks contact the glue and become trapped by the glue.
2. The method according to claim 1, further comprising folding the support to provide a housing having inside surfaces defining a protective enclosure and at least one side opening, the glue being on at least one inside surface, and the housing being placed at a location surface so that ticks walk from the location surface through the at least one side opening directly into the protective enclosure and contact the glue.
3. The method according to claim 2, wherein the glue is present at least on a bottom inside surface at a bottom of the housing.
4. The method according to claim 2, further comprising removing the tick attractant from a separate package and applying the tick attractant to the glue or the housing.
5. The method according to claim 2, wherein the support is formed from a biodegradable material and the method further comprises placing the trap in an environment where the trap will be exposed to moisture and the trap decomposing when exposed to the moisture.
6. The method according to claim 5, wherein the trap is colored to blend in with the environment.
7. The method according to claim 1, further comprising providing a housing having at least one side opening and inside surfaces defining a protective enclosure, the support having the glue thereon is constructed to be removably inserted within the protective enclosure, and placing the housing having the support inserted therein at a location surface so that ticks walk from the location surface through the at least one side opening directly into the protective enclosure and contact the glue.
8. The method according to claim 7, wherein the glue is at a bottom of the housing.
9. The method according to claim 1, wherein the location is in a dwelling.
10. The method according to claim 1, wherein the location is in a place where cloths are stored.
11. The method according to claim 1, further comprising placing a plurality of the tick traps around an area to provide a tick reduced protection area.
12. The method according to claim 1, wherein the trap is free of ingredients that are toxic, poisonous, or harmful to humans or mammals so that when the trap decomposes the environment will not be contaminated with a toxic, poisonous or harmful material.
13. The method according to claim 1, wherein the support is formed from a biodegradable material.
14. The method according to claim 13, wherein the biodegradable material contains a water resistant coating so that the trap can be utilized in the environment for at least a week before degrading.
15. The method according to claim 13, further comprising placing a plurality of traps in an area to trap ticks and allowing the traps to decompose.
16. The method according to claim 1, wherein the plant based scent comprises an extract from a part of a basil plant.
17. The method according to claim 1, wherein the plant based scent comprises an essential oil of basil.
18. The method according to claim 1, wherein the tick attractant is on the glue or incorporated into the glue.
19. A method of trapping ticks comprising: folding a support to provide a housing having at least

one side opening and inside surfaces defining a protective enclosure, a glue being on at least one inside surface, the glue is constructed to trap ticks touching the glue by sticking to the ticks, a tick attractant is incorporated in the glue, is on the glue, is on the support, or is on the housing, and the tick attractant comprising a plant based scent formed from basil; placing the housing on a location surface to trap ticks; and the plant based scent attracting ticks so that ticks walk from the location surface through the at least one side opening directly into the protective enclosure and contact the glue, wherein the glue traps the ticks.

20. The method according to claim 19, further comprising applying the plant based scent to the glue or the support.

21. The method according to claim 19, wherein the trap is free of ingredients that are toxic, poisonous, or harmful to humans or mammals so that when the trap decomposes the environment will not be contaminated with a toxic, poisonous or harmful material.

22. The method according to claim 19, wherein the support comprises a biodegradable material.

23. The method according to claim 22, wherein the biodegradable material comprises cellulose.

24. The method according to claim 22, further comprising placing a plurality of traps in an area to trap ticks and allowing the traps to decompose.

25. The method according to claim 24, wherein the traps are colored to blend into the area.

26. The method according to claim 19, further comprising placing a plurality of traps in an area to trap ticks.

27. The method according to claim 19, wherein the plant based scent comprises an extract from a part of a basil plant.

28. The method according to claim 19, wherein the plant based scent comprises an essential oil of basil.

29. The method according to claim 19, wherein the tick attractant is on the glue or incorporated into the glue.

30. A method of trapping ticks comprising: providing a housing having at least one side opening and inside surfaces defining a protective enclosure; providing a support constructed to be removably inserted into the housing, a glue being on the support, the glue is constructed to trap ticks touching the glue by sticking to the ticks, a tick attractant is incorporated in the glue, is on the glue, is on the support, or is on the housing and the tick attractant comprising a plant based scent formed from basil; placing the housing on a location surface to trap ticks; and the plant based scent attracting ticks so that ticks walk from the location surface through the at least one side opening directly into the protective enclosure and contact the glue, wherein the glue traps the ticks.

31. The method according to claim 30, further comprising applying the plant based scent to the glue or the support.

32. The method according to claim 30, wherein the trap is free of ingredients that are toxic, poisonous, or harmful to humans or mammals so that when the trap decomposes the environment will not be contaminated with a toxic, poisonous or harmful material.

33. The method according to claim 30, wherein the plant based scent comprises an extract from a part of a basil plant.

34. The method according to claim 30, wherein the plant based scent comprises an essential oil of basil.

35. The method according to claim 30, wherein the tick attractant is on the glue or incorporated into the glue.
